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DATA SCIENCE WITH MACHINE LEARNING & ARTIFICIAL INTELLIGENCE

Become Proficient in Python, Numpy, Pandas, Matplotlib, Seaborn, Sklearn, Machine Learning, Deep Learning, Computer Vision, NLP









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Career Opportunities In DATA SCIENCE with Machine Learning & AI

Data Scientist

Analyzes data to extract insights and build predictive models for informed decision-making.

Machine Learning Engineer

Develops and deploys machine learning models for real-world applications.

Data Analyst

Gathers, cleans, and visualizes data to provide actionable insights for businesses.

Business Intelligence Analyst

Creates data reports and visualizations to monitor and inform business performance.

Data Engineer

Manages data collection, cleaning, and transformation for analysis.

AI Researcher

Advances artificial intelligence by developing new algorithms and models.

NLP Engineer

Develops language algorithms for chatbots, sentiment analysis, and translation.

Python Developer

Codes, tests, and maintains software, specializing in web development, data analysis, or automation.

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ABOUT US

UpFlairs is an innovative educational technology company with a clear mission to elevate the skills and employability of students throughout India.

Our dedicated team is committed to fostering the next generation of tech talent, equipping them with cutting-edge skills in emerging technologies and has educated more than 47K students all over the globe including IITs, NITS, Deemed Universities and other colleges.

We offer the courses that are most trending technologies of the recent era in the fields like AI/ML, Data Science, Cloud Computing, DevOps, Full Stack Web Development, Embedded Systems, IoT, and Robotics. These courses are meticulously designed to provide students with the practical skills and knowledge required to excel in tech-driven careers, making them not just job-ready but industry leaders of tomorrow.

We are not only limited to training the youth of the country but also provide Lab setups to various Colleges and Universities and Project solutions to other companies for the domains like AI-ML, IoT, ROBOTICS AND CLOUD.

EDUCATION



PROJECT SOLUTION



LAB SETUP



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Who Should Go for This **DATA SCIENCE WITH MACHINE LEARNING & AI?**



College Student

Start shaping your career right from college time by learning in-demand skills like Data Science with Machine Learning & AI.



Graduate

Time for you to upskill, prepare yourself for a bright future, and kickstart a career in Data Science with Machine Learning & AI.



College Dropout/Job Seeker

Not sure which career is right for you? Not finding any jobs? Learn Data Science with Machine Learning & AI with us to land your first job.



Freelancer

Whether you're already a freelance Data Scientist or looking to launch your career in freelancing, expertise in Machine Learning & AI is the most lucrative skill set for freelancers in the data science domain.



Looking to Switch Career

Not satisfied with your current job profile? Switch to Machine Learning & AI, an easy-to-learn, most impressive, and highly-paid skill.

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Why Learn DATA SCIENCE WITH MACHINE LEARNING & AI With Upflairs Only?

Trusted by Learners

Our training and support system intrigue learners.

Expert Trainers

Professional Trainers having experience of delivering more than 30k students. We ensure the high quality training always aimed at career building.

Completely Practical-Oriented

We offer 100% practical training with regular assessment, assignment and projects.

Industry-Recognized Certification

The certificate received by the learners on course completion is valid nationally and internationally. You can easily share , add it to your resume and explore great opportunities.

Online & Offline Batches

You can upskill yourself from anywhere by joining our live training batches
Also we provide classroom training in Jaipur.

Career Assistance

We help you build a great career as a developer. Our team offers you career guidance, pepares you for interview, and assists you in finding job.

Hands-On Projects

You will get to work on multiple projects during the training so that you can gain some experience.

Comprehensive Curriculum

The course includes most comprehensive Curriculum, covering all breadths and depths in detail.

DATA SCIENCE WITH MACHINE LEARNING		
Duration – 6 Months (240 Hours)		
1	**Introduction (3 Hours)**	• Introduction with AI & Machine Learning • Data Science Vs Data Engineering Vs Data Analysis vs AI • Use of Data in the world of Data Science and machine learning • Connecting with Upflairs Community • Basic Linux/Windows Commands • Setting Up GITHUB & Google Colab/Kaggle
2	**Core Python Overview (30 Hours)**	• Python and Vs Code Download and Installation setup. • Python introduction, compiler vs interpreter, program execution • Keywords, Data Types, Operators and its types. • Conditional statements, Looping, exception Handling file handling • Comprehensions • Python User Defined Functions • Generators, iterators, decorators • Lambda Expressions • Python Modules: Usage and Installation • Creating own modules, and packages • Built In modules os, time ,datetime, re, random, • The file handling in python • Dealing with Excel/CSV/txt files
3	**Oops python (11 Hour)**	• Class, object, instance, constructor, methods, abstraction, • Inheritance and its types • encapsulation • inheritance, polymorphism
	Mini Projects (4 Hours)	***Project 1:*** • ***ATM simulation project*** • ***Bulk Email sender*** • ***BMI calculator***
	Capstone Project(2 Hours)	***Capstone Project 1:*** • ***CLI Based Chat Application***
4	**Git & Github (3 hours)**	• What is GitHub? • Understanding repositories, commits, and branches. • Creating a GitHub account. • Creating a new repository. • Cloning a repository. • Making commits. • Pushing changes to GitHub. • Pulling changes from a repository. • Creating and switching branches. • Merging branches. • Resolving merge conflicts. • Forking a repository. • Creating pull requests. • Reviewing and merging pull requests.

5	**Web frameworks Flask and streamlit (8 Hours)**	• Familiarity with HTML and CSS. • What is Flask? • Setting up a Flask environment. • Creating your first Flask app ("Hello, World!"). • Understanding routes and URLs. • Dynamic routes and URL parameters. • Using route decorators • Introduction to Jinja2 templating. • Rendering HTML templates. • Passing data from Flask to templates. • Creating and processing HTML forms. • Handling POST and GET requests. • Validating form data. • Structuring a Flask project. • Implementing basic CRUD operations. • Testing the application locally. • Introduction to SQLAlchemy or sqlite3 or mysql. • Setting up a database connection. • Performing CRUD operations with SQLAlchemy or sqlite3 or mysql.
	Assignment	***Assignment 1:*** ***Build portfolio website using flask***
6	**Web Scrapping (4 Hours)**	• Understanding web scraping fundamentals. • Explore Beautiful soop, auto scraper python library for web scraping. • Scraping the live Data from flipkart, git hub , and from Wikipedia
7	**Database (MYSQL + MongoDB) (16 Hours)**	• Understanding the Database and DBMS, and RDBMS, • Understanding Sql and Nosql database. • Working with Sql database. • Writing CURD operations in database. • Write affective queries to filter out the data from the database. • perform various type of joins
8	**Numpy in Python (3 Hour)**	• What is numpy? • Why numpy is introduced on over the list? • Difference between numpy and python list. • Creating numpy array 1D, 2D, 3D Array. • Accessing item from array using indexing, • Data Manipulation in Array. • Searching and filtering in Numpy array. • Function to generate quick numpy arrays like (zero, ones, linspace, random) • Aggregation functions in numpy. • Understand the difference between argmin and argmax. • Understand the difference between argsort and sort

9	**Pandas (7 Hours)**	• What is pandas? • Overview of pandas • Installation of pandas • Data manipulation • Data transformation • Data cleaning • Missing value imputation • Redundant value handling. • Sorting, filtering • Understand about Series and Dataframe • Series: Creating, Indexing, and Basic Operations • DataFrame: Creating, Indexing, and Basic Operations • DataFrame vs. Series: Differences and Use Cases • Reading and Writing CSV, Excel, JSON, and other file formats • Functions in pandas (head(), tail(), info(), describe(),isnull(), dropna(), fillna(),loc[], iloc[],sort_values(), sort_index(),groupby(),agg(),concat(),apply(),map(),merge(),lambda
10	**Data Visualization (6 Hours)**	• Data is Beautiful…!!! • Visualization Libraries in Python • MATPLOTLIB PYPLOT: line, scatter, pie, box, area etc • Decorating the plots using Matplotlib (labels, colors, markers, legend, grids, figure sizes etc) • The Subplots and axes in matplotlib; Showing Images • Pandas Visualization: Basic Plots • bar, barh, hist, box, kde, density, area, scatter, pie plots • Plotting with Missing Data • Easy and advanced Data Visualization from Seaborn
11	**Statistics (12 Hours)**	• Learn about different statistical types, from descriptive to inferential statistics. • Understand measures of central tendency (mean, median, mode) and dispersion (variance, standard deviation, range). • Correlation, covariance, association, 5 number summary. • Permutation and combination. • Grasp the concepts of probability, percentiles, quartiles, and the normal distribution. • Z-score, central limit theorem, and left/right-skewed distributions.
	Mini Project (3 Hours)	***Project 3 :*** • ***"End to End Data Analytics project on real world dataset"***

12	**Machine Learning Supervised (20 Hours)**	• Understanding the concept of Machine Learning • Types of Learning and their sub-categories • Linear Regression, Polynomial Regression • Regression model evaluation metrics. • L1 & L2 Regularization • Logistic Regression • Performance metrics (model evaluation techniques) • Support Vector Machines (SVM) • Kernel Nearest Neighbors (KNN) • Decision Trees Classifier • Hyper parameter Turning. • Bagging:- Random Forest Classifier • Boosting:- Ada Boost,Xgboost • Naive Bayes with Bayes theorem
	Mini Project (15 Hours)	***Project 4 :*** • ***Email spam & ham classification*** • ***Laptop price prediction (Assignment)*** • ***Old bike price prediction*** • ***Disease prediction***
	Assignment	***Assignment 2:*** ***Bangalore house price prediction*** ***Insurance premium prediction***
13	**Machine Learning Unsupervised (8 Hours)**	• K-means & K-means++ • Hierarchical clustering (agglomerative, Divisive) • DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
	Mini Project (3 Hours)	***Project 5:*** • ***Customer Segmentation***
	Capstone Project (5 Hours)	***Capstone Project 2:*** • ***Farmer Guider Ai app.***
14	**Image Processing (6 Hours)**	• What is image processing? • Applications of image processing • Image Representation: Pixels and bit-depth, Grayscale vs. Color Images. • Introduction to OpenCV, PIL (Pillow) • Loading, displaying, and saving images. • Image attributes (size, shape, and data type) • Scaling, translation, rotation, and affine transformation. • Flipping, resizing, and cropping images. • RGB to Grayscale. • Concept of histograms. • Histogram equalization and its applications. • Erosion, Dilation, • object shape filtering, object border shape detection.

15	**Computer Vision with CNN (10 Hours)**	• what is computer vision • real world use cases of computer vision. • Understanding CNN working process • Filters, activation function, pooling layers, • flatten layers, cnn architecture, image feature extraction, • CNN architecture implementation. • Image classification. • Transfer learning techniques • Object detection: Localization using yolo.
	Mini Project (10 Hours)	***Project 6 :*** • ***Dog cat image classification*** • ***image search project*** • ***Sign language detection***
	Assignment	***Assignment 3:*** ***Leaf Disease detection yolo*** ***Flower image classification***
16	**Natural Language Processing (4 Hours)**	• What is NLP? Linguistic to Natural Language! • NLTK in Python for Text Processing • Text to Speech and Speech to Text Modules in Python • Optical character recognition (OCR): Text recognition. • Generating Word Clouds
17	**Natural Language Processing with RNN (12 Hours)**	• What is RNN • How RNN is different from ANN • Types of RNN • Text cleaning steps, stemming and lemmatization, tokenization, stop words, pos tagging. Bag of words, tfidf, embedding layer. • The Concept of Long-Short-Term Memory (LSTM) • LSTM based Neural Networks for Future Prediction!! • GRU.
18	**Generative Ai (4 Hours)**	• What is Generative AI. • Real world application of Generative Ai. • Encoder and Decoder Based Architecture understanding, • Building Language Translator using encoder and decoder based architecture using LSTM • What is LLM models? • How the LLM model works, • Learn about LLM model working process, and generate the data. • Transformers Architecture understanding • BERT and GPT model architecture • NER extraction and text summarization using pre-trained model • Named entity recognition (NER extraction)

	Minor project (11 Hours)	***Project 7:*** • ***Stock price prediction*** • ***Language translation*** • ***NER extraction and Text summarization using pre-trained model*** • ***Text generation Ai model from scratch***
19	**AWS (4 Hours)**	• Overview of AWS services • Launching an EC2 instance. • Connecting to EC2 via SSH. • Setting up a Python environment on EC2. • Preparing a machine learning model for deployment. • Clone project on EC2 machine from github. • Deploying the model on EC2. • Exposing the model via a simple Flask API.
	Capstone Project (8 hours)	***Capstone Project 3:*** • **Build AI Job finder application** • **Generative Ai based Chabot**

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TRAINING COMPLETION CERTIFICATE

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CERTIFICATE
OF PARTICIPATION

Proudly Presented to

Student Name

X Sem |Department of

from X College has successfully completed 6 Months Internship & Training Program in **Full Stack Web Development** from D /M / Y to D /M / Y at **Upflairs Pvt. Ltd.**

Siddharth Singh
Founder & Director
Upflairs Pvt. Ltd.

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Ministry of MSME, Govt. of India

#startupindia

Enrollment No. : UF/XXX/XXXX

Trainer Name
Designation
Upflairs Pvt. Ltd.

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OUR STUDENTS FEEDBACK

I'm very happiest to share my experience about my summer internship at Upflair was an incredible learning experience in the data science domain. The hands-on projects, supportive mentors, and real-world applications made it the perfect opportunity for college students to develop their skills. The team's dedication to helping us grow both technically and professionally truly stood out, making it a highly rewarding experience and it is really good to learn something valuable.

Ritu Saini
PIET, Jaipur

I had a good experience at learning in upflairs. We had regular classes and practical implementation of machine learning and data science projects....
Faculty is best as educator and also teaches everything in beginner friendly manner.
We did a many projects and practically learnt the deployment.

Riya Sharma
Arka Jain University

I recently completed my Python internship at Upflairs. The program was well-structured, offering a solid blend of theoretical knowledge and hands-on projects. The mentors were incredibly supportive, providing guidance whenever needed and encouraging creativity and problem-solving.

Sanjay kumar
Arya College of Engineering

It was a great and amazing experience in upflairs pvt ltd, In my 45 days intership of data science and machine learning & AI I have learnt a lot and it helped in making a strong base for data science and ML & AI .

Manvik
PIET, Jaipur

I Anshika Mundra from Poornima Institute of Engineering and Technology have completed my summer training cum internship of 15 days in Frontend Web Development Course. My experience here was fabulous .The knowledgeable and understanding instructors helped me gain various technical and practical skills. I worked on various projects, applying my knowledge to various tasks.

Anshika Mundra
PIET, Jaipur

Great learning experience at Upflairs . Gained valuable skills, worked on meaningful projects, and enjoyed a supportive team environment. Highly recommend for hands-on experience and professional growth. Minor challenges, but overall positive.

Hotri Shrimali
Amity University, Rajasthan

Hello everyone, My name is Antish kumar from career point university. I had enrolled in 2 months Data sciences internship + traning program. The staff is so friendly, helpful to students and resolve all doubts. Our sir Ranjit singh assist me or whole class very well.

Antish Kumar
Career Point University, Kota

It was a really great experience in learning about domain and all the concept of Full stack web development. Environment and level of teaching skills were really up to the mark. Overall, it was a really good experience.

Ajit Prajapat
PIET, Jaipur

I completed my 45-days internship with Er. Ranjit Singh in Upflairs in ye domain of data science with AI&Ml... Here I got a Vry gud xperience with full practical implementation in AI&ML .... The faculty taught all the concepts very well...including the coding part ..... I did multiple projects under his guidance and also developed my final project in the same domain .

Akash Mishra
Arka Jain University



Banasthali Vidyapeeth







Glimpse of Events

Upflairs



NIT, Jalandhar





JECRC, Jaipur



Banasthali Vidyapeeth

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From Startups & SMBs to Brands & MNCs, companies are actively looking for skilled **Data Science with Machine Learning & AI Engineer** at lucrative packages.

Grab these opportunities by holding mastery of **Data Science with Machine Learning & AI!**





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